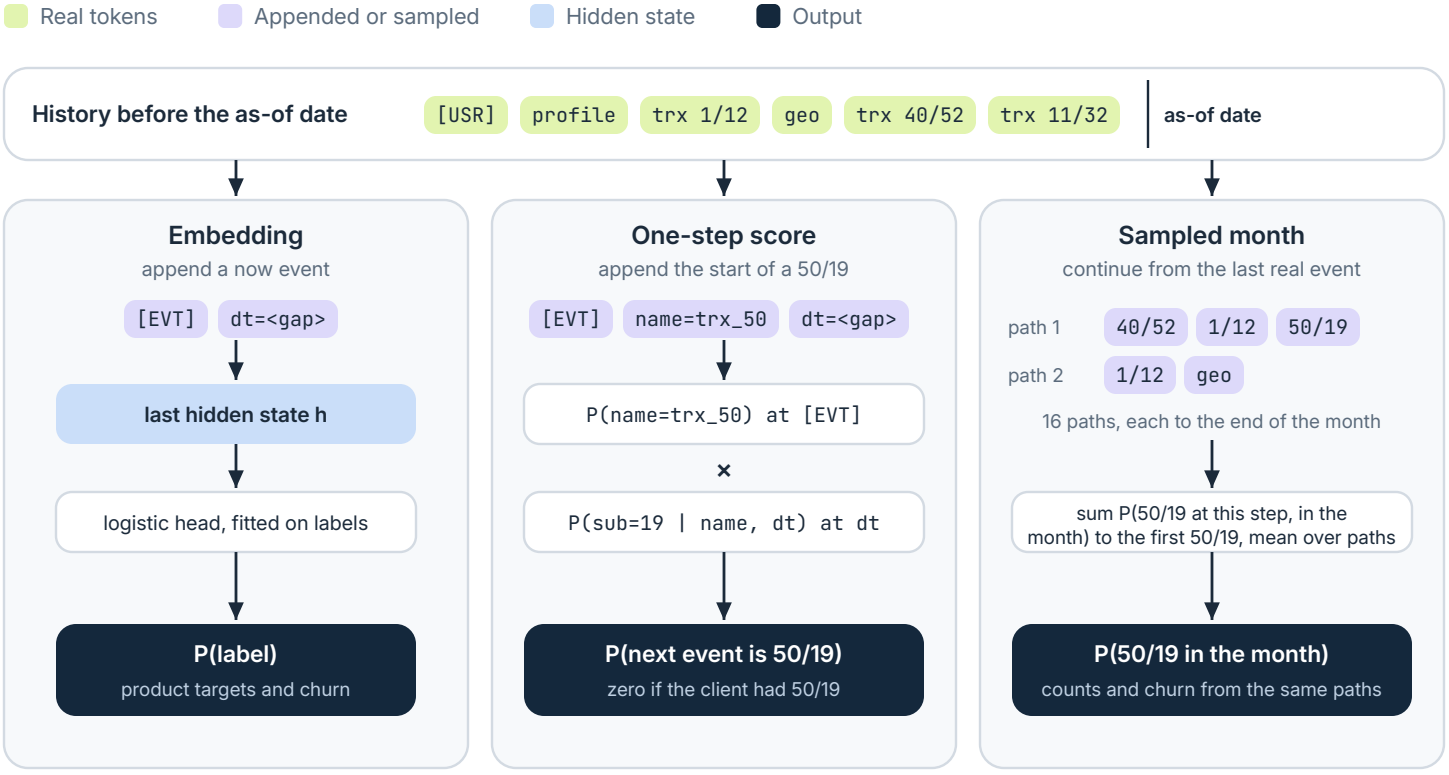
Real tokens
Appended or sampled
Hidden state
Output
History before the as-of date
[USR]
profile
trx 1/12
geo
trx 40/52
trx 11/32
as-of date
Embedding
append a now event
[EVT]
dt=<gap>
last hidden state h
logistic head, fitted on labels
P(label)
product targets and churn
One-step score
append the start of a 50/19
[EVT]
name=trx_50
dt=<gap>
P(name=trx_50) at [EVT]
×
P(sub=19 | name, dt) at dt
P(next event is 50/19)
zero if the client had 50/19
Sampled month
continue from the last real event
path 1
40/52
1/12
50/19
path 2
1/12
geo
16 paths, each to the end of the month
sum P(50/19 at this step, in the month) to the first 50/19, mean over paths
P(50/19 in the month)
counts and churn from the same paths